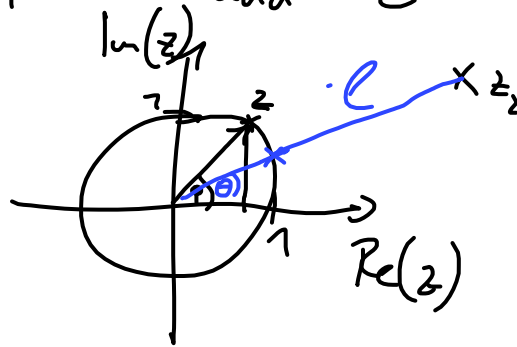


Komplexe Zahlen $\mathbb{C}$

Es sei $z \in \mathbb{C}$, dann gilt $z = a + bj$ mit $j = \sqrt{-1}$

→ wegen EULER-Identität

$$e^{j\pi} = -1 \quad \text{und} \quad e^{j\varphi} = \cos(\varphi) + j \sin(\varphi)$$

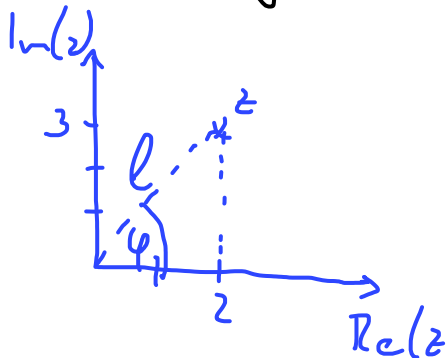


$$\begin{aligned} \ell e^{j\varphi} &= \ell (\cos(\varphi) + j \sin(\varphi)) \\ &= \ell \cos(\varphi) + \ell j \sin(\varphi) \end{aligned}$$

Transformation zwischen Typen

① Summenform → e-Form

$$z = 2 + 3j$$



$$\Rightarrow m = \tan(\varphi)$$

$$\tan^{-1}(m) = \varphi, \quad \text{wobei} \quad m = \frac{\Delta \operatorname{Im}(z)}{\Delta \operatorname{Re}(z)}$$

$$\Rightarrow \tan^{-1}\left(\frac{3}{2}\right) = \varphi$$

$$\Rightarrow \underline{\varphi = 0,98} \quad (\hat{=} 56,3^\circ)$$

$$\rho = \sqrt{\operatorname{Re}(z)^2 + \operatorname{Im}(z)^2}$$

$$\rho = \sqrt{2^2 + 3^2}$$

$$\underline{\rho = \sqrt{13}}$$

$$z = 2 + 3j = \underline{\underline{\sqrt{13} \cdot e^{0,98j}}}$$

② e-Form $\rightarrow$ Summenform

$$z = \underset{\varphi}{\rho} e^{j \left(\frac{\pi}{4} \right)}$$

$\operatorname{rad}(\varphi)$

$$z = 3 \left(\cos\left(\frac{\pi}{4}\right) + j \sin\left(\frac{\pi}{4}\right) \right)$$

$$z = 3 \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} j \right)$$

$$z = \frac{3\sqrt{2}}{2} + \frac{3\sqrt{2}}{2} j$$